

WEEK 3 – PASSWORD CRACKING PROJECT

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INTERNSHIP BATCH: SEP26 BATCH -B083

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1. Project Overview

Week 3 focused on understanding password-cracking techniques in an authorized cybersecurity lab environment. The project involved learning how password-protected files can be converted into crackable hash formats and how password-recovery tools can be used with wordlists to identify the correct password.

The main activities included password cracking using John the Ripper (JTR) and password cracking using NetworkWalks training tools.

2. Objectives

- Understand the basic concept and workflow of password cracking.
- Learn how to obtain a crackable hash from a password-protected PDF.
- Prepare and use a wordlist for password recovery.
- Perform password recovery using John the Ripper.
- Use NetworkWalks tools for hash generation and password testing.
- Verify the results and document the completion of the lab activities.

3. Tools and Technologies Used

- Kali Linux
- John the Ripper (JTR)
- RockYou Wordlist
- NetworkWalks Hash Calculator
- NetworkWalks Password Cracker
- Password-protected PDF files

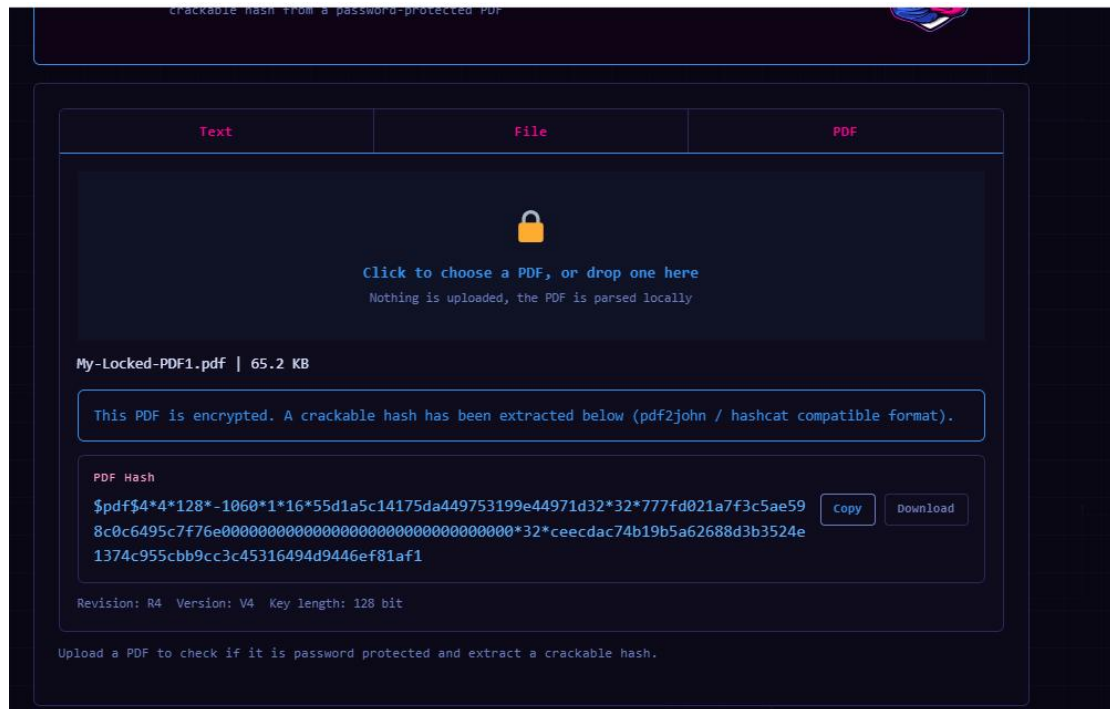
4. Module 1 – Password Cracking Using John the Ripper

The first activity involved working with password-protected PDF files in the authorized training environment. The PDF files were converted into a suitable hash format using the NetworkWalks Hash Calculator.

After obtaining the hash, the RockYou wordlist available in Kali Linux was used with John the Ripper. JTR tested different password candidates from the wordlist until a matching password was identified.

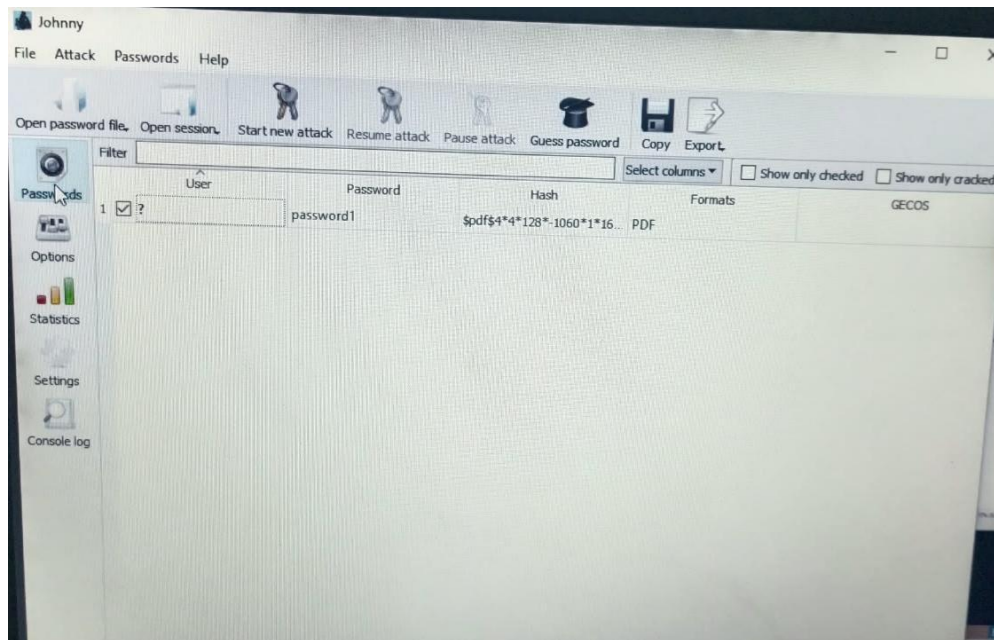
4.1 Hash Extraction

The password-protected PDF file was provided to the NetworkWalks Hash Calculator. The tool generated a crackable hash that could be used for password-recovery testing.



4.2 John the Ripper Execution

PDF 1,PDF 2,PDF 3 was tested using John the Ripper through the Johnny graphical interface in a controlled cyber security lab environment. The password-protected PDF was loaded into Johnny, and the tool identified the PDF hash and performed the password-cracking process. The attack was completed successfully, and the recovered password was displayed as password1 in the Password column. This practical demonstrated how weak and predictable passwords can be recovered using password-cracking tools.

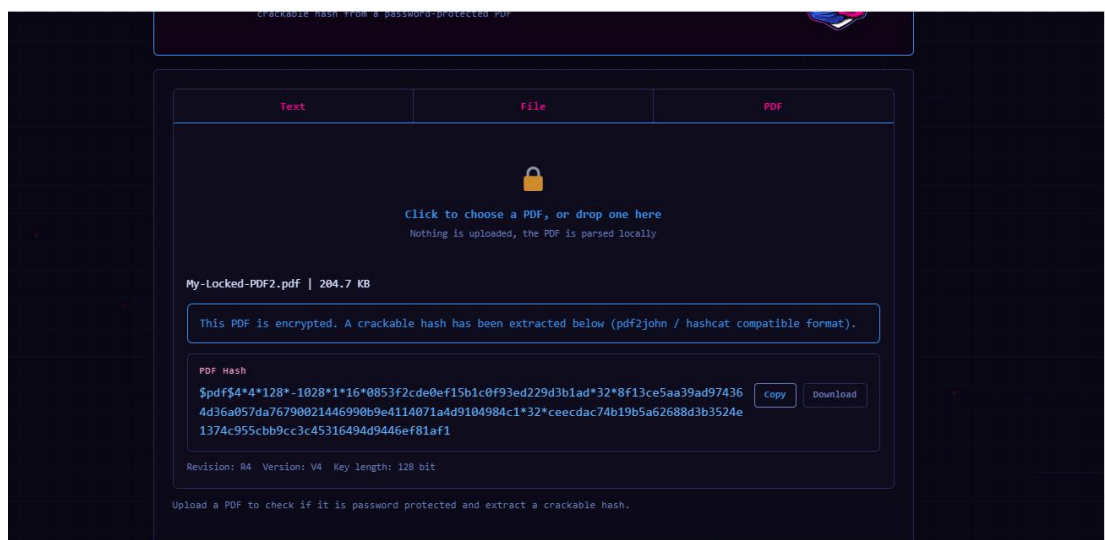


5. Module 2 – Password Cracking Using NetworkWalks Tools

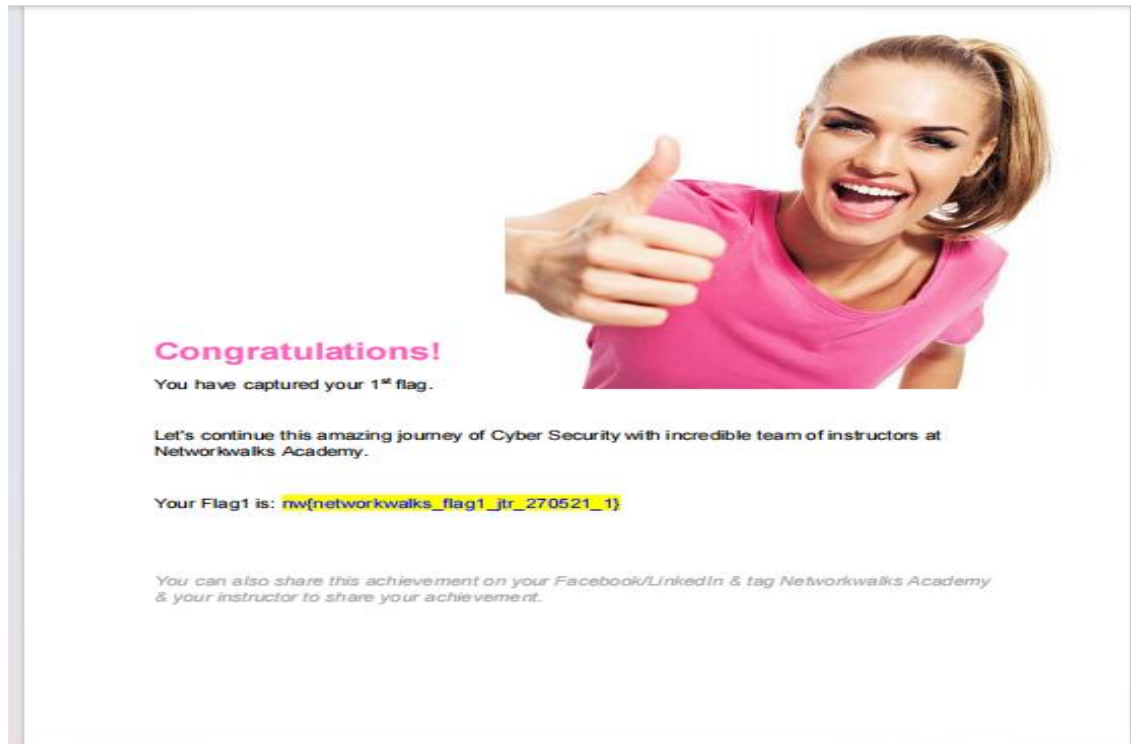
The second activity involved using the NetworkWalks training tools. The password-protected PDF was processed through the hash calculator to generate a suitable hash.

The generated hash was then provided to the NetworkWalks Password Cracker. The tool performed the password-testing process and displayed the result after finding a matching password.

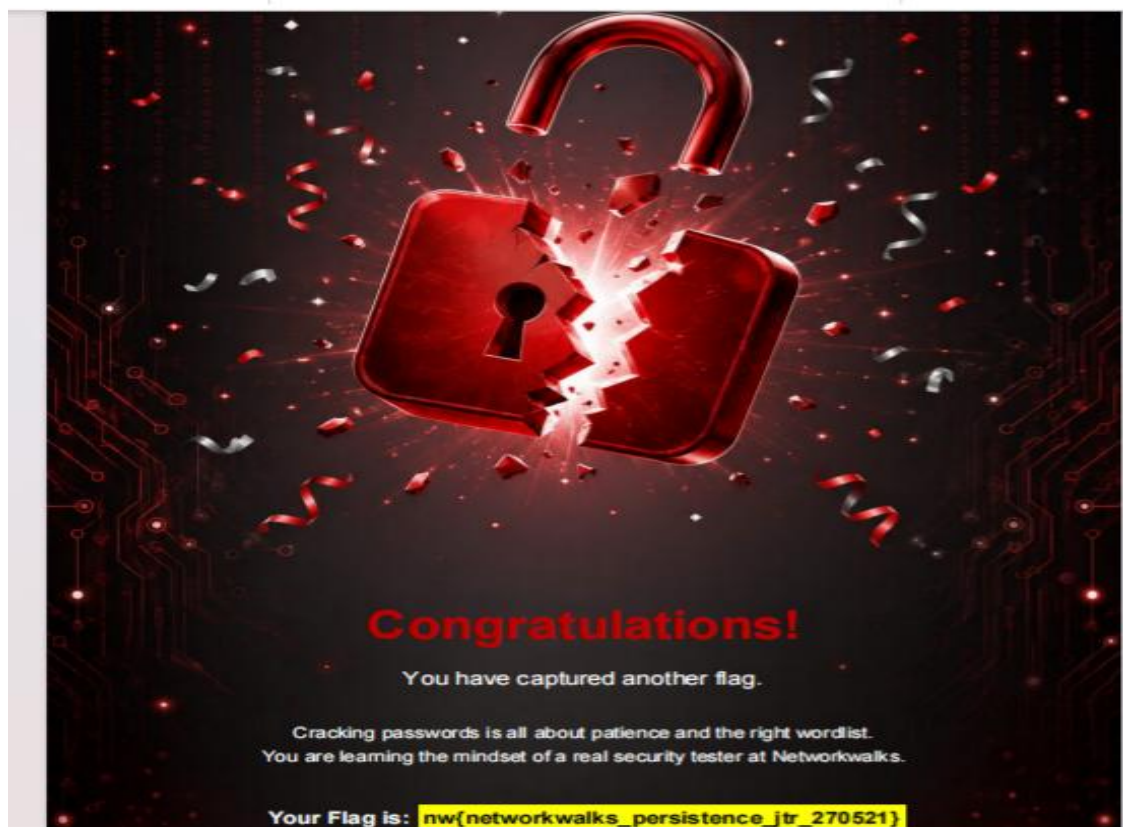
My locked pdf1 hash



My locked Pdf 1 password cracked



FLAG 2



FLAG 3



Congratulations!

You have captured your flag.

Let's continue this amazing journey of Cybersecurity with incredible team of experts at Networkwalks.

7. Learning Outcomes

Through this project, I learned:

- How password-protected files are associated with password hashes.
- How hashes can be extracted for authorized password-recovery testing.
- How John the Ripper works with wordlists.
- How the RockYou wordlist can be used to test password candidates.
- How to interpret password-cracking results.
- How to use NetworkWalks tools for practical cybersecurity exercises.
- The importance of documenting screenshots and results during a cybersecurity lab.

8. Security and Ethical Considerations

Password cracking should only be performed on files, systems, or accounts for which explicit authorization has been provided.

This project was performed in an authorized cybersecurity training environment using intentionally supplied laboratory files. Real credentials, private password hashes, or unauthorized systems should never be tested.

9. Conclusion

The Week 3 project provided practical experience with password-recovery techniques using John the Ripper and NetworkWalks tools. The activities helped me understand the process of extracting a password hash, using a wordlist, testing password candidates, and verifying successful results in an authorized cybersecurity laboratory environment..